INSTITUT FOR MATEMATIK
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Statistical Models
Week 10: Markov Chain Monte Carlo: the Gibbs sampler

> ! There are no lectures on Wednesday, November 5th.
>
> Instead, we will have a longer lecture on Friday, November 7th lectures within 10:15-13:45, with a lunch break of at least half an hour. Friday lecture will be in Aud D1.
> In course week 10, there will only be lectures on Friday, November 7th. Next week, lectures on Wednesday are extended. Since the exercise class is moved from Tuesday to Wednesday you will have a free Tuesday early morning.

This week's only lecture concentrates on the *Gibbs sampler*, a method that allows to generate random samples from the posterior distribution a multidimensional parameter. In essence it consists in updating one element of the multidimensional parameter at a time.

The other very important MCMC method is the *Metropolis-Hastings algorithm*, that enables us to simulate from a distribution that is known up to a constant. If time allows, we will start with the Metropolis algorithm in the end of Friday's lecture.

In order to understand the algorithms, you will implement them yourselves in R. Later in this course you will use software "off the shelf" to run MCMC. You will learn to use JAGS and the R package `rjags` to analyze simple Bayesian models.

Install JAGS

In the coming weeks, we will use the R package `rjags` for Bayesian analysis. `rjags` is an interface to the software `JAGS` (Just Another Gibbs Sampler) that you need to install separately on your computer (not from within R). Please install it from `sourceforge` (link).

Relevant Readings

- Notes on Markov chains with finite and continuous state space,
- Gibbs sampler: Reich & Ghosh, Section 3.2, 3.2.1.

Lecture Material

- Link to onenotes on Google drive
- Gibbs sampler for bivariate normal distribution
- Listing 3.2 from Reich & Ghosh for experimentation
- Proof of the Gibbs sampler

Exercises

TODO Eine aufgabe weniger

Problem 10.1: MAP and numerical integration

Reich & Ghosh, Chapter 3, Exercise 2

Assume that $Y \mid \mu \overset{\text{indep.}}{\sim} \text{Normal}(\mu, \sigma_i^2)$, for $i = 1, \ldots, n$, with σ_i known and improper prior distribution $\pi(\mu) = 1$ for all μ.

a. Give a formula for the MAP estimator for μ.

b. We observe $n = 3, Y_1 = 12, Y_2 = 10, Y_3 = 22$, and know that $\sigma_1 = \sigma_2 = 3$ and $\sigma_3 = 10$. Compute the MAP estimate of μ.

c. Use numerical integration to compute the posterior mean of μ.

d. Plot the posterior distribution of μ and indicate the MAP and the posterior mean estimates on the plot.

Hints

Parts a) and b) can be solved without R; you might want to compare your analytical result for the MAPin b) with numerical results by R function `optim`.
For parts c) and d), find ideas in listing 3.1, R&G page 72. You can use function `adaptIntegrate` from package `cubature`, or just work with base R function `integrate` here. You can download the R code here.
Both functions calculate the integral of a function that can take a vector of all arguments and would return a vector of all function values. If your own function only returns a single number, you will get weird results. The remedy is then to create a new version of your function by applying R function `Vectorize`.

Problem 10.2: Bayesian CLT approximation

Reich & Ghosh, Chapter 3, Exercise 3.

Assume $Y_i \mid \lambda \overset{\text{indep}}{\sim} \text{Poisson}(N_i\lambda)$ for $i = 1, \ldots, n$.

a. Identify a conjugate prior for λ and derive the posterior that follows from this prior.

b. Using the prior $\lambda \sim \text{Uniform}(0, 20)$, derive the MAP estimator of λ.

c. Using the prior $\lambda \sim \text{Uniform}(0, 20)$, plot the posterior on a grid of λ assuming $n = 2, N_1 = 50, N_2 = 100, Y_1 = 12$, and $Y_2 = 25$ and demonstrate that the MAP estimate is indeed the maximizer (draw a vertical line).

d. Use the Bayesian CLT to approximate the posterior of λ under the setting of (c), plot the approximate posterior, and compare the plot with the plot from (c).

Hint

For d), find code for Figure 3.2 (R&G) on the course supplement page

Problem 10.3: Inference for the variance in a hierarchical normal model

Reich & Ghosh, Chapter 3, Exercise 4 (a) – (d).

Consider the model $Y_i \mid \sigma_i^2 \overset{\text{indep}}{\sim} \text{Normal}(0, \sigma_i^2)$ for $i = 1 \ldots n$, where $\sigma_i^2 \mid b \sim \text{InvGamma}(a, b)$ and $b \sim \text{Gamma}(1, 1)$.

a. Derive the full conditional posterior distributions for σ_1^2 and b.

b. Write pseudocode for Gibbs sampling, i.e., describe in detail each step of the Gibbs sampling algorithm.

c. Write your own Gibbs sampling code (not in JAGS) and plot the marginal posterior density for each parameter. Assume $n = 10$, $a = 10$ and $Y_i = i$ for $i = 1, \dots 10$.

d. Repeat the analysis with $a = 1$ and comment on the convergence of the MCMC chain.

% You can skip JAGS question e. e. Implement this model in (c) using JAGS and compare the results with the results in (c).

Problem 10.4: Gibbs sampling for comparison of two groups in normal model

Reich & Ghosh, Chapter 3, Exercise 5 (a) – (c).
Consider the model $Y_i \mid \mu.\sigma^2 \sim \text{Normal}(\mu, \sigma^2)$ for $i = 1, \dots, n$ and $Y_i \mid \mu, \sigma^2 \sim \text{Normal}(\mu + \delta, \sigma^2)$ for $i = m + 1, \dots, n + m$, where $\mu, \delta \sim \text{Normal}(0, 100^2)$ and $\sigma^2 \sim \text{InvGamma}(0.01, 0.01)$.

a. Give an example of a real experiment for which this would be an appropriate model.

b. Derive the full conditional posterior distributions for μ, δ, and σ^2.

c. Simulate a dataset from this model with $n = m = 50$, $\mu = 10, \delta = 14$, and $\sigma = 2$. Write your own Gibbs sampling code (not in JAGS) to fit the model above to the simulated data and plot the marginal posterior density for each parameter. Are you able to recover the true values reasonably well? Try to find out if your result in (c) varies a lot when you simulate again.

%You can skip JAGS question d. (d) Implement this model using JAGS and compare the results with the results in (c).

Problem 10.5: Interesting and relevant properties of the AR(1) process

In the lectures, we discuss(ed) the autoregressive process AR(1) as an example for a Markov chain (see the interactive experiment)
Consider the AR(1) chain $X_0, X_1, \dots$ given by

$$X_n = \phi X_{n-1} + \lambda + \tau Z_n, \quad n = 1, 2, \dots,$$

where

$$Z_1, Z_2, \dots \overset{\text{i.i.d.}}{\sim} \text{N}(0, 1), \ -1 < \phi < 1, \ \lambda \in \mathbb{R}, \ \tau > 0,$$

and Z_n is independent of X_m for all n and $m < n$.

a. Give an argument, that the X_n in such a chain must have a normal distribution. Thus, the stationary distribution must be a normal distribution.

Show that the normal distribution with mean $\mu = \lambda/(1 - \phi)$ and variance $\sigma^2 = \tau^2/(1 - \phi^2)$ is the stationary distribution of the chain.

Run some simulation(s) to confirm this result.

b. Confirm by calculation or simulation, that, for all $n \in \mathbb{N}$ and $k = 0, 1, \dots, n - 1$

$$\text{Cov}(X_n, X_{n-k}) = \phi^k \sigma^2, \quad \text{or, equivalently,} \quad \text{Cor}(X_n, X_{n-k}) = \phi^k.$$

c. Investigate the variance of the sample mean

$$\overline{X}_{(m)} = \frac{1}{m} \sum_{i=1}^{m} X_i$$

from values $X_1, \ldots, X_m$ from the AR(1) chain started in a random value X_0 from the stationary distribution, and compare with the variance that you would get when the samples $X_1, \ldots, X_m$ were drawn *independently* from the stationary distribution.

Do this by simulating for several values of ϕ, also for some negative ones. What can you conclude?